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Project: Inferential Statistics: Hypothesis Testing

Topic 1: Inferential Statistics: Population Proportion: One Sample

Dataset: President and Opponent Heights (cm)

Source:

<https://www.statcrunch.com/app/?dlim=comma&dataurl=https%3A%2F%2Ftdx.acs.pearson.com%2FGetPlayerFile.ashx%3Fguid%3D2418bf61-dfd7-450a-8d1f-ce6b97430571&ft=true>

Description: The dataset I shall use includes the heights of 20 randomly selected presidents and the heights of their opponents.

I shall treat the data as a sample, and I shall find the proportion of presidents who were taller than their opponents.



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MyStatLab Data Set

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Row	PRESIDENT	HEIGHT	HEIGHT OPP	
1	T. Roosevelt	178	175	
2	Truman	175	173	
3	Van Buren	168	180	
4	Jefferson	189	170	
5	Clinton	188	188	
6	McKinley	170	178	
7	Johnson	192	180	
8	Taylor	173	174	
9	Pierce	178	196	
10	Taft	182	178	
11	Jackson	185	171	
12	F. Roosevelt	188	182	
13	Nixon	182	180	
14	Hayes	173	178	
15	Polk	173	185	
16	Hoover	182	180	
17	Harding	183	178	
18	Coolidge	178	180	
19	Eisenhower	179	178	
20	Reagan	185	177	

The presidents that were taller than their opponent are: T. Roosevelt, Truman, Jefferson, Johnson, Taft, Jackson, F. Roosevelt, Nixon, Hoover, Harding, Eisenhower, and Reagan.

12 presidents were taller than their opponents. This is the number of successes.

We define success as a president who is taller than the opponent.

Sample size, $n = 20$

Number of successes, $x = 12$

Number of failures $= 20 - 12 = 8$

Objectives:

- (1.) Construct a 95% confidence interval estimate of the population proportion of presidents who were taller than their opponents.
- (2.) Interpret the confidence interval.
- (3.) Based on the result, does it appear that greater height is an advantage for presidential candidates? Why or why not?

Parameter to estimate: Population Proportion

Test: prop-test

Reason for the Test: We were not given the population standard deviation.

Verify Requirements for the Test.

- (1.) The sample is a simple random sample.
- (2.) The selection of the presidents in the sample are independent.
- (3.) The probability for the selection of each president is the same.
- (4.) There are at least 12 successes and 8 failures.

Data Analysis Using R Language

```
# Load the dataset
presidentsData <- read.csv("Presidents.csv")

# Identify the number of presidents taller than their opponents
# Note: R replaces spaces in column names with dots (HEIGHT.OPP)
tallerThanOpponent <- sum(presidentsData$HEIGHT > presidentsData$HEIGHT.OPP)
totalPresidents <- nrow(presidentsData)

# Calculate the sample proportion
phat <- tallerThanOpponent / totalPresidents

# Construct a 95% confidence interval for the population proportion
# Using the standard normal approximation (Wald interval)
alpha <- 0.05
zCritical <- qnorm(1 - alpha/2)
standardError <- sqrt((phat * (1 - phat)) / totalPresidents)

lowerBound <- phat - (zCritical * standardError)
upperBound <- phat + (zCritical * standardError)

# Output the results
cat("Number of presidents taller than opponents:", tallerThanOpponent, "\n")
cat("Sample Proportion (p-hat):", phat, "\n")
cat("95% Confidence Interval:", lowerBound, "to", upperBound, "\n")
cat("In percentages:", lowerBound * 100, "% to", upperBound * 100, "%\n")
```

```
Number of presidents taller than opponents: 12
Sample Proportion (p-hat): 0.6
95% Confidence Interval: 0.3852967 to 0.8147033
In percentages: 38.52967 % to 81.47033 %
```

Interpretation: I am 95% confident that the population of presidents who were taller than their opponents is approximately between 38.5% and 81.5% (rounded to 1 decimal place).

Topic 2: Hypothesis Testing: Testing a Claim About Population Mean

Dataset: Systolic Blood Pressure Levels (in mmHg)

Source:

<https://www.statcrunch.com/app/?dlim=comma&ft=false&dataurl=https%3a%2f%2ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3dcd3e0b2f-71a7-4abc-9d4d-24da3eaaa361#>

Description: The dataset is a list of 100 systolic blood pressure levels in mmHg.



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Row	SystolicBloodPressureLevels
1	106
2	118
3	140
4	106
5	108
6	126
7	88
8	118
9	122
10	94
11	112
12	140
13	100
14	122
15	158
16	118
17	122
18	128
19	112
20	122

Row	SystolicBloodPressureLevels
21	132
22	112
23	126
24	134
25	122
26	124
27	162
28	128
29	134
30	140
31	116
32	130
33	158
34	100
35	114
36	168
37	114
38	128
39	132
40	114

Row	SystolicBloodPressureLevels
41	122
42	132
43	116
44	122
45	130
46	120
47	142
48	146
49	116
50	128
51	138
52	96
53	112
54	142
55	92
56	114
57	122
58	104
59	118
60	144

Row	SystolicBloodPressureLevels
61	126
62	128
63	134
64	96
65	106
66	116
67	112
68	114
69	132
70	100
71	118
72	124
73	100
74	136
75	156
76	106
77	120
78	120
79	108
80	122

Row	SystolicBloodPressureLevels
81	134
82	108
83	112
84	120
85	120
86	128
87	130
88	108
89	132
90	164
91	120
92	134
93	136
94	110
95	120
96	130
97	94
98	106
99	126
100	114

Variable: Systolic Blood Pressure Level

Systolic blood pressure levels above 120 mm Hg are considered to be high. For the 100 systolic blood pressure levels listed in the accompanying data set, the mean is 122.24000 mm Hg and the standard deviation is 15.93745 mm Hg. Assume that a simple random sample has been selected. Use a 0.01 significance level to test the claim that the sample is from a population with a mean greater than 120 mm Hg.

Objectives:

- (1.) Test the claim that the 100 systolic blood pressure levels are from a population with a mean greater than 120 mm Hg using the Critical Value Method.
- (2.) Test the claim that the 100 systolic blood pressure levels are from a population with a mean greater than 120 mm Hg using the Probability Value Method.
- (3.) State the decision.
- (4.) State the conclusion.
- (5.) State the interpretation.

Verify Requirements for the Test.

- (1.) The sample is a simple random sample.
- (2.) The sample size is at least 30, justifying normality based on the Central Limit Theorem.

Null Hypothesis: $\mu = 120 \text{ mm Hg}$

Alternative Hypothesis: $\mu > 120 \text{ mm Hg}$

Test: One-sample, Right-tailed

Data Analysis Using R Language

```
▶ # Load the dataset
data <- read.csv("SystolicBloodPressureLevels.csv")
# Given statistics
sampleSize <- 100
meanLevel <- 122.24
standardDeviationLevel <- 15.93745
populationMean <- 120
alpha <- 0.01

# Calculate the test statistic (t)
testStatistic <-
  (meanLevel - populationMean) / (standardDeviationLevel / sqrt(sampleSize))

# Calculate the Degrees of Freedom
df <- sampleSize - 1

# 1. Critical Value Approach
# Find the critical value for a right-tailed test at alpha = 0.01
testCritical <- qt(1 - alpha, df)
```



```
# 2. P-value Approach
# Find the p-value for the calculated testStatistic
pValue <- pt(testStatistic, df, lower.tail = FALSE)

# Output results
cat("Test Statistic (t):", testStatistic, "\n")
cat("Critical Value:", testCritical, "\n")
cat("P-value:", pValue, "\n")

# Decision Logic
if (testStatistic > testCritical) {
  cat("Decision: Reject the null hypothesis.\n")
} else {
  cat("Decision: Fail to reject the null hypothesis.\n")
}
```

```
... Test Statistic (t): 1.405495
Critical Value: 2.364606
P-value: 0.08150237
Decision: Fail to reject the null hypothesis.
```

Critical Value Method:

Critical z – value = 2.364606

Test Statistic = 1.405495

$1.405495 < 2.364606$

The test statistic is less than the critical z-value. This means that it does not fall in the critical region.

Decision: Fail to reject the null hypothesis.

Probability Value Method:

$$P - value = 0.08150237$$

$$significance\ level = 0.01$$

$$0.08150237 > 0.01$$

The probability value is greater than the significance level.

Decision: Do not reject the null hypothesis.

Conclusion: There is not sufficient evidence at the 0.01 significance level to support the claim that the sample is from a population with a mean systolic blood pressure level greater than 120 mm Hg.

Interpretation: The population mean for the Systolic Blood Pressure Level is not significantly more than 120 mm Hg.

References (MLA 9)

“StatCrunch.” *Statcrunch.com*, 2026,

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